

Lab 4: VibeCoding — Tic-Tac-Toe

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1. Objective

Use Vibe Coding tools to fix the broken starter code and add required features in under 3 to 4 attempts. Ensure clean separation of concerns, move validation, persistent scoreboard across rounds, and starting player toggle.

2. Initial Bug Analysis

- **Diagonal Wins Ignored:** `check_winner()` only inspected rows and columns.
- **Win on 9th Move Overwritten by Draw:** `is_board_full()` was checked prior to `check_winner()`.
- **Clicks Accepted Post-Game:** No `round_over` guard in `handle_click()`.
- **No Occupancy Check:** Players could overwrite already occupied cells.

3. Prompt Engineering & Resolution (3 Iterations)

Attempt 1: Fix Win & Draw Detection (Task 1)

Prompt: Fix the win/draw detection: diagonals must count as wins, a win on the 9th move must not be reported as a draw, and clicks after game over must be ignored.

Fix: Updated `rules.py` to include both diagonals in `check_winner()`. Reordered `check_round_end()` in `game_engine.py` to check winner before board-full. Added `if self.round_over: return` to `handle_click()`.

Attempt 2: Move Validation (Task 3)

Prompt: Reject clicks on already-occupied cells so that players cannot overwrite moves, keeping board and turn state intact.

Fix: In `handle_click()`, added `if self.board[row][col] is not None: return`.

Attempt 3: Scoreboard & Restart Controls (Tasks 2 & 4)

Prompt: Implement persistent scoreboard across rounds (X wins, O wins, Draws). Provide [R] for new round, [M] for full match reset, and [T] to toggle starter. If O starts, computer makes the first move.

Fix: Added `self.scores`, `new_round()`, `reset_match()`, and `toggle_starter()`. Updated `renderer.py` to draw real-time scoreboard, starter indicator, and controls.

4. Automated Unit Test Verification

```
All 7 test cases in test_game.py executed successfully:
test_diagonal_wins: OK | test_win_on_last_move_is_not_draw: OK | test_genuine_draw: OK |
test_occupied_cell_move_rejected: OK | test_no_moves_accepted_after_round_over: OK |
test_persistent_scoreboard: OK | test_starting_player_choice_and_computer_first_move: OK
Output: Ran 7 tests in 0.001s - OK
```

5. Deliverables Summary

1. **Updated Code:** `tic-tac-toe/game/` and `test_game.py`
2. **Videos:** Before video (bug reproduction) & After video (fixed game)
3. **Chat Documentation:** Exported as DOCX and PDF.